Intro to Data Science - HW 7

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```
# Enter your name here: Sofia Quintero
```

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```
# 1. I did this homework by myself, with help from the book and the professor.
```

In our last assignment, we explored data visualization in R using the **ggplot2** package. This homework continues to use ggplot2, but this time, with **maps**. Additionally, we will merge datasets using the built-in **merge()** function, which provides a similar capability to a JOIN in SQL (don't worry if you do not know SQL). Many analytical strategies require joining data from different sources based on a key - a field that two datasets have in common.

Step 1: Load the data

A. Some people may struggle to eat a healthy diet due to a lack of access to supermarkets, grocery stores, or other sources of healthy and affordable food. This limited availability to food is commonly referred to as **food insecurity**. However, as it is frequently linked to deep-rooted systemic injustices, many agree that the term "food apartheid" is more fitting.

There are numerous methods for measuring food store access and determining which areas are low-access neighborhoods with few healthy food options. In this homework assignment, we will focus on accessibility to sources of healthy food, as measured by the distance to a store in a given area.

The dataset available at https://data-science-intro.s3.us-east-2.amazonaws.com/FoodInsecurity.csv includes a variable labeled LAPOP1_10 that indicates the count of individuals residing more than 1 mile (for urban areas) or 10 miles (for rural areas) away from a supermarket.

To analyze the data, you can read it into a dataframe named **insecurity_df** and visually inspect its contents.

Please note in a comment what geographic area an observation (a row in the dataframe) represents - whether it pertains to a single city, a state, or another region.

Finally, use a command to display the number of observations and variables in insecurity_df.

```
library("tidyverse")
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
```

```
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
insecurity_df <- read_csv("https://data-science-intro.s3.us-east-2.amazonaws.com/FoodInsecurity.csv")
```

```
## Rows: 3142 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): State, County, AveragePovertyRate, MedianFamilyIncome, Largest_city...
## dbl (2): Pop2010, LAPOP1_10
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(insecurity_df)
```

```
## # A tibble: 6 x 9
##   State   County         Pop2010 LAPOP1_10 AveragePovertyRate MedianFamilyIncome
##   <chr>   <chr>            <dbl>     <dbl> <chr>              <chr>
## 1 Alabama Autauga County   54571    18503. 16.13078591        69337.5
## 2 Alabama Baldwin County  182265    45789. 11.84554563        72665.74194
## 3 Alabama Barbour County   27457     5634. 29.29932484        44792.44444
## 4 Alabama Bibb County      22915      365. 12.19352439        60645.5
## 5 Alabama Blount County    57322     3902. 14.85074758        60437.66667
## 6 Alabama Bullock County   10914     7480. 30.65326247        39524.66667
## # i 3 more variables: Largest_city <chr>, city_state <chr>, abbr <chr>
```

```r
# Comment: Each observation represents a county in a state, including
# information on that county's largest city
```

```r
glimpse(insecurity_df)
```

```
## Rows: 3,142
## Columns: 9
## $ State              <chr> "Alabama", "Alabama", "Alabama", "Alabama", "Alabam~
## $ County             <chr> "Autauga County", "Baldwin County", "Barbour County~
## $ Pop2010            <dbl> 54571, 182265, 27457, 22915, 57322, 10914, 20947, 1~
## $ LAPOP1_10          <dbl> 18503.2255, 45789.0465, 5634.3628, 365.1407, 3902.2~
## $ AveragePovertyRate <chr> "16.13078591", "11.84554563", "29.29932484", "12.19~
## $ MedianFamilyIncome <chr> "69337.5", "72665.74194", "44792.44444", "60645.5",~
## $ Largest_city       <chr> "Prattville", "Daphne", "Eufaula", "Brent", "Oneont~
## $ city_state         <chr> "Prattville, Alabama", "Daphne, Alabama", "Eufaula,~
## $ abbr               <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL~
```

B. Calculate the average of MedianFamilyIncome in insecurity_df. Explain in a comment why the mean() function might not work and how to fix it.

```r
mean(as.numeric(insecurity_df$MedianFamilyIncome))
```

```
## Warning in mean(as.numeric(insecurity_df$MedianFamilyIncome)): NAs introduced
## by coercion

## [1] NA
```

```r
# mean() won't work because of the NA introduced by coercion warning
```

```r
which(is.na(as.numeric(insecurity_df$MedianFamilyIncome)) != is.na(insecurity_df$MedianFamilyIncome))
```

```
## Warning in which(is.na(as.numeric(insecurity_df$MedianFamilyIncome)) != : NAs
## introduced by coercion

## [1] 1817 2413
```

```r
insecurity_df[c(1817, 2413), 6]
```

```
## # A tibble: 2 x 1
##   MedianFamilyIncome
##   <chr>
## 1 #DIV/0!
## 2 #DIV/0!
```

```r
# the values in these rows can't be converted to numeric
```

```r
# this can be fixed by replacing the values with 0
insecurity_df$MedianFamilyIncome <- gsub("#DIV/0!", "0", insecurity_df$MedianFamilyIncome)
```

```r
mean(as.numeric(insecurity_df$MedianFamilyIncome))
```

```
## [1] 63933.25
```

C. What is the population of the most populous county in the dataframe? Which state is it in?

```r
max(insecurity_df$Pop2010)  #population of most populous county
```

```
## [1] 9818605
```

```r
insecurity_df[which.max(insecurity_df$Pop2010), 1]  #it's in California
```

```
## # A tibble: 1 x 1
##   State
##   <chr>
## 1 California
```

D. Without a reference point, it is difficult to understand the significance of the values in **LAPOP1_10** (remember, this is the variable that shows the number of people living too far from a store and thus considered at risk of food insecurity). Create a new column in insecurity_df called **insecurity_ratio** that represents the ratio of LAPOP1_10 to Pop2010. Explain what it means in a comment.

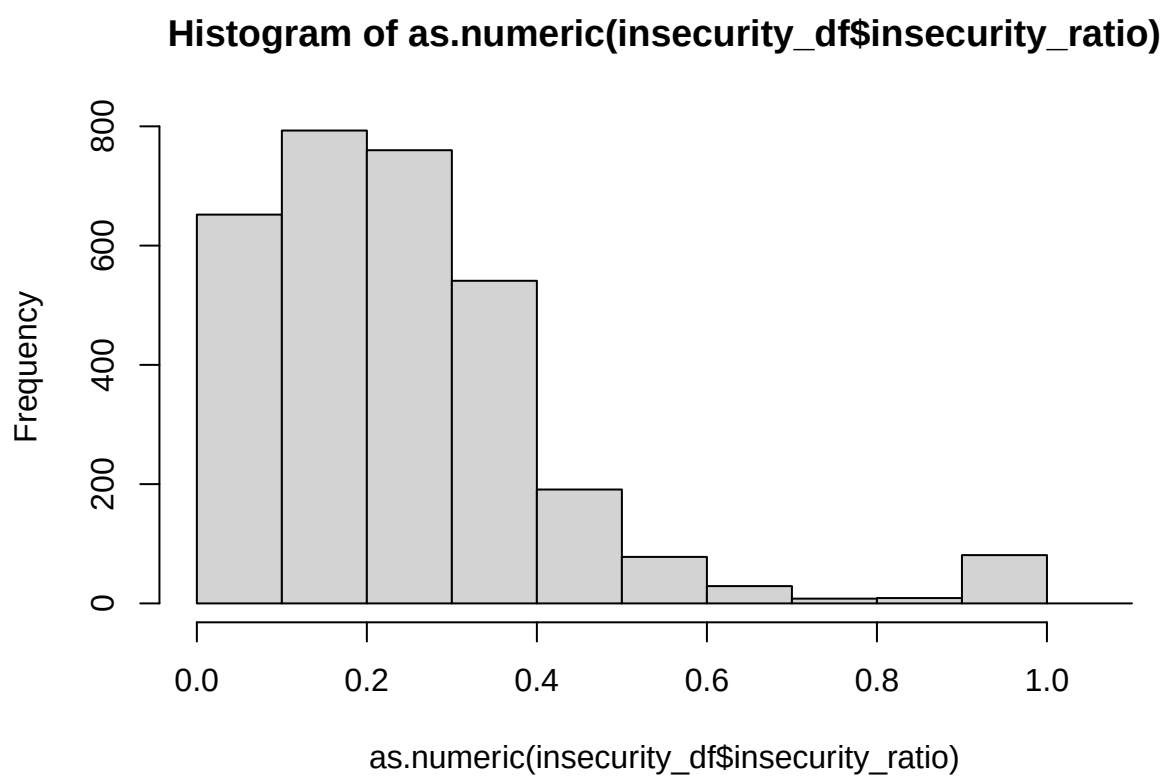
```r
insecurity_df$insecurity_ratio <- (insecurity_df$LAPOP1_10/insecurity_df$Pop2010)
# for how many people in the population, how many live too far from a store
```

E. Provide descriptive statistics for the insecurity ratio (min, max, mean, and median) using a single function. Create a histogram with ggplot2 to visualize the distribution.

```r
summary(insecurity_df$insecurity_ratio)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.1193  0.2147  0.2439  0.3237  1.0000
```

```r
hist(as.numeric(insecurity_df$insecurity_ratio))
```



Step 2: Merge the food apartheid data with the city data

A. Read the JSON file at https://intro-datascience.s3.us-east-2.amazonaws.com/cities.json into a dataframe called **population_df**.

Use a function to display the variable data types in population_df and add comments explaining what each variable contains.

```r
options(repos = c(CRAN = "https://cran.r-project.org"))
install.packages("RCurl")
```

```
##
## The downloaded binary packages are in
##  /var/folders/rm/s4vv52kd55g34xzsxxqn9p2c0000gn/T//Rtmp8sGTRe/downloaded_packages
```

```r
install.packages("jsonlite")
```

```
##
## The downloaded binary packages are in
##  /var/folders/rm/s4vv52kd55g34xzsxxqn9p2c0000gn/T//Rtmp8sGTRe/downloaded_packages
```

```r
library(RCurl)
```

```
##
## Attaching package: 'RCurl'

## The following object is masked from 'package:tidyr':
##
##     complete
```

```r
library(jsonlite)
```

```
##
## Attaching package: 'jsonlite'

## The following object is masked from 'package:purrr':
##
##     flatten
```

```r
dataset <- getURL("https://intro-datascience.s3.us-east-2.amazonaws.com/cities.json",
    ssl.verifypeer = FALSE)
population_df <- jsonlite::fromJSON(dataset)
```

```r
glimpse(population_df)
```

```
## Rows: 1,000
## Columns: 7
## $ city                     <chr> "New York", "Los Angeles", "Chicago", "Housto~
## $ growth_from_2000_to_2013 <chr> "4.8%", "4.8%", "-6.1%", "11.0%", "2.6%", "14~
## $ latitude                 <dbl> 40.71278, 34.05223, 41.87811, 29.76043, 39.95~
## $ longitude                <dbl> -74.00594, -118.24368, -87.62980, -95.36980, ~
## $ population               <chr> "8405837", "3884307", "2718782", "2195914", "~
## $ rank                     <chr> "1", "2", "3", "4", "5", "6", "7", "8", "9", ~
## $ state                    <chr> "New York", "California", "Illinois", "Texas"~
```

```r
# city- city name (chr) growth_from_2000_to_2013- percent growth between 2000
# and 2013 (chr) latitude- latitude of city (dbl) longitude- longitude of city
# (dbl) population- population of city (chr) rank- rank in population count
# (chr) state- state city is in (chr)
```

B. To merge insecurity_df and pop_df, we need a common column. We'll use the **city_state variable** in insecurity_df, which we'll create by joining the city and state columns in population_df. Explain the code provided:

```r
population_df$city_state <- paste0(population_df$city, ", ", population_df$state)

# Comment: city_state column is modified by converting city name, a comma, and
# state into a string through the paste0() function
```

C. Merge the two dataframes using the city_state column from both dataframes and store the resulting dataframe in **merged_df**.

```r
merged_df <- merge(insecurity_df, population_df, by = "city_state")
```

D. Display the dimensions (number of observations and variables) of merged_df and comment on the number of variables.

```r
dim(merged_df)
```

```
## [1] 423  17
```

```r
# The 10 variables from insecurity_df The 8 variables from population_df One
# variable overlaps (city_state) making the new column 17 variables
```

Step 3: Visualize the data

A. Plot points for each city in merged_df on a map of the United States. Use color to represent the insecurity ratio.

```r
library(ggplot2)
library(maps)
```

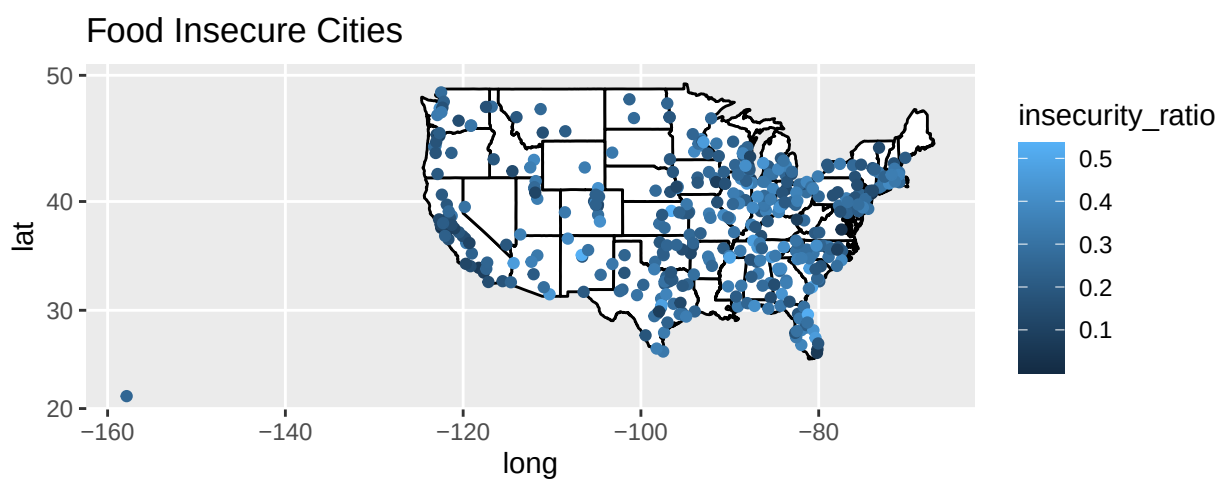
```
##
## Attaching package: 'maps'

## The following object is masked from 'package:purrr':
##
##     map
```

```r
library(ggmap)
```

```
## i Google's Terms of Service: <https://mapsplatform.google.com>
##   Stadia Maps' Terms of Service: <https://stadiamaps.com/terms-of-service/>
##   OpenStreetMap's Tile Usage Policy: <https://operations.osmfoundation.org/policies/tiles/>
## i Please cite ggmap if you use it! Use `citation("ggmap")` for details.
```

```r
library(mapproj)
us <- map_data("state")
us$state_name <- tolower(us$region)
map1 <- ggplot(us, aes(map_id = state_name))
map1 <- map1 + aes(x = long, y = lat, group = group) + geom_polygon(fill = "white",
    color = "black")
map1 <- map1 + expand_limits(x = merged_df$longitude, y = merged_df$latitude)
map1 <- map1 + coord_map() + ggtitle("Food Insecure Cities")
plottedMap <- map1 + geom_point(data = merged_df, aes(x = longitude, y = latitude,
    color = insecurity_ratio), inherit.aes = FALSE)
plottedMap
```



B. The map you just created is probably not perfect; comment briefly on what could be improved.

```r
# Comment: The map could be bigger so it becomes clearer what cities the points
# pertain to, or the size of the points could be smaller
```

Step 4: Group by State

A. Create a dataframe of state-by-state average **insecurity_ratio** using **group_by()** and **summarize()**. Save the result in **state_insecurity_df**.

```r
state_insecurity_df <- merged_df %>%
    group_by(merged_df$State) %>%
    summarize(ratio_avg = as.numeric(mean(insecurity_ratio)))
```

B. Identify the most and least food-insecure states in state_insecurity_df and their food insecurity ratios. Show the code you used to determine them.

```r
state_insecurity_df[which.max(state_insecurity_df$ratio_avg), ]
```

```
## # A tibble: 1 x 2
##   `merged_df$State` ratio_avg
##   <chr>                 <dbl>
## 1 Wyoming               0.382
```

```r
# Wyoming is the most food-insecure state with a max avg of 0.3823024
state_insecurity_df[which.min(state_insecurity_df$ratio_avg), ]
```

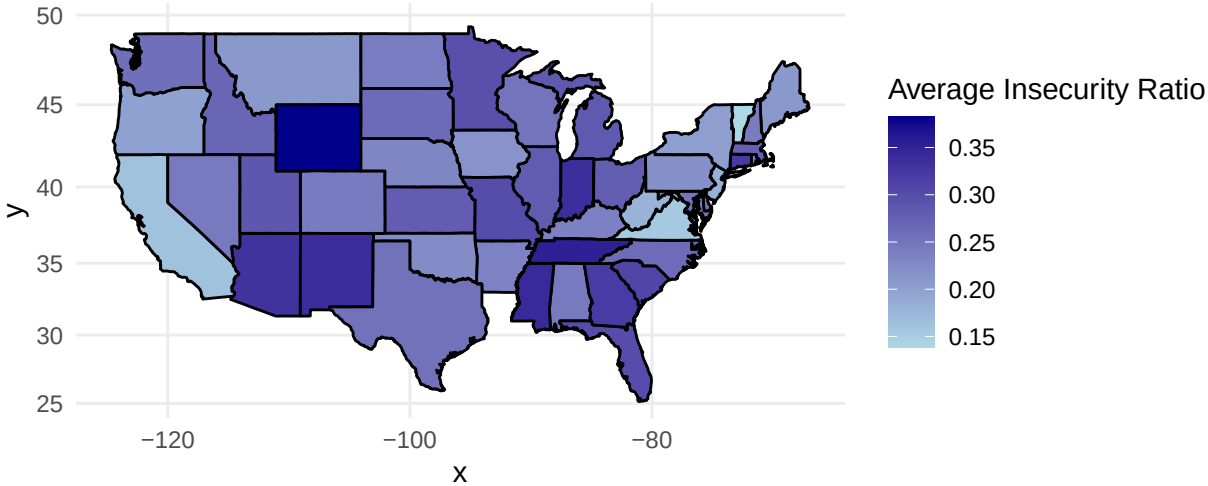
```
## # A tibble: 1 x 2
##   `merged_df$State` ratio_avg
##   <chr>                 <dbl>
## 1 Vermont               0.139
```

```r
# Vermont is the lest food-insecure state with a avg ratio of 0.139224
```

Step 5: Create a map of the United States, with the color of each state representing the average food insecurity ratio in state_insecurity_df

A. Ensure you correctly set the map limits and use coord_map appropriately, and make sure that lighter shades represent lower food insecurity.

```r
state_insecurity_df$state <- tolower(state_insecurity_df$`merged_df$State`)
ggplot(state_insecurity_df, aes(map_id = state)) + geom_map(map = us, aes(fill = ratio_avg),
    color = "black") + expand_limits(x = us$long, y = us$lat) + coord_map() + scale_fill_gradient(low =
    high = "darkblue", name = "Average Insecurity Ratio") + theme_minimal()
```



Step 6: Data Analysis

A. Explain in a comment what insights you can glean about food insecurity from the maps you created in this homework.

```
# Comment: The plotted cities show what specific cities may be struggling with
# food insecurity as well as to what degree. This allows for a closer look as
# to what specific areas and neighborhoods may need reform. The states shaded
# by ratio provide a broader insight into what larger areas may need attention,
# especially when comparing the cities. For example, California has many
# plotted cities but still returns a low insecurity ratio.  Whereas Wyoming has
# a low number of plotted cities but a high insecurity ratio, indicating that
# city is worth checking out.
```